What DM_{host} and the host properties correlate with (whole sample)

Pair	Sample	N	r_S (p)	AD p	bootstrap 95%	delete-3: frac. p<0.05	partial τ | z (p)	DM_{cosmic} MC r_S (frac.; AD)	BH q	BH q (AD)
Redshift selection										
log M* vs z	All	51	+0.32 (0.024)	0.022	[+0.05, +0.55]	0.83	–	–	0.12	0.19
log SFR vs z	All	46	+0.34 (0.020)	0.090	[+0.05, +0.58]	0.87	–	–	0.12	0.23
log sSFR vs z	All	46	+0.14 (0.35)	0.96	[-0.14, +0.40]	0.00	–	–	0.54	1.00
R_e vs z	All	55	+0.05 (0.73)	0.99	[-0.27, +0.35]	0.00	–	–	0.80	1.00
offset (kpc) vs z	All	45	+0.30 (0.042)	0.036	[-0.02, +0.59]	0.52	–	–	0.16	0.19
DM_host (observer frame) vs z	All	55	-0.10 (0.47)	0.37	[-0.38, +0.19]	0.00	–	–	0.67	0.59
DM_host vs z	All	55	+0.08 (0.55)	0.92	[-0.21, +0.36]	0.00	–	+0.03 (0.00; 0.01)	0.73	1.00
Stars and star formation										
DM_host vs log M*	All	51	-0.01 (0.96)	0.91	[-0.29, +0.28]	0.00	-0.01 (0.91)	-0.03 (0.00; 0.00)	0.97	1.00
DM_host vs log SFR	All	46	+0.24 (0.10)	0.021	[-0.08, +0.53]	0.09	+0.19 (0.067)	+0.19 (0.07; 0.23)	0.24	0.19
DM_host vs log SFR	z < 0.2	20	+0.01 (0.97)	0.50	[-0.45, +0.48]	0.00	-0.01 (0.95)	-0.06 (0.00; 0.00)	0.97	0.75
DM_host vs log SFR	z ≥ 0.2	26	+0.55 (0.004)	0.050	[+0.16, +0.80]	0.99	+0.40 (0.005)	+0.46 (0.71; 0.34)	0.07	0.19
DM_host vs log SFR	Prospector SFRs	30	+0.11 (0.56)	0.13	[-0.25, +0.48]	0.00	+0.11 (0.35)	+0.07 (0.00; 0.02)	0.73	0.26
DM_host vs log SFR	Prospector, z ≥ 0.2	17	+0.48 (0.051)	0.032	[-0.02, +0.83]	0.21	+0.32 (0.082)	+0.41 (0.29; 0.33)	0.17	0.19
DM_host vs log sSFR	All	46	+0.28 (0.064)	0.063	[-0.02, +0.53]	0.24	+0.18 (0.048)	+0.23 (0.21; 0.23)	0.18	0.21
DM_host vs log sSFR	z ≥ 0.2	26	+0.35 (0.084)	0.049	[-0.07, +0.67]	0.15	+0.23 (0.052)	+0.30 (0.22; 0.35)	0.22	0.19
DM_host vs log Sigma_SFR	All	46	+0.34 (0.022)	0.085	[+0.02, +0.61]	0.81	+0.26 (0.023)	+0.28 (0.47; 0.19)	0.12	0.23
DM_host vs log Sigma_SFR	z ≥ 0.2	26	+0.50 (0.009)	0.076	[+0.09, +0.79]	0.89	+0.37 (0.012)	+0.45 (0.67; 0.26)	0.11	0.23
DM_host vs log Sigma_SFR	Prospector SFRs	30	+0.28 (0.13)	0.33	[-0.10, +0.61]	0.07	+0.21 (0.10)	+0.23 (0.07; 0.03)	0.27	0.59
DM_host vs log Sigma_SFR	literature R_e	36	+0.28 (0.092)	0.68	[-0.07, +0.59]	0.13	+0.22 (0.078)	+0.26 (0.23; 0.00)	0.22	0.84
DM_host vs Delta MS	All	46	+0.31 (0.036)	0.043	[+0.02, +0.56]	0.54	+0.22 (0.022)	+0.26 (0.32; 0.30)	0.15	0.19
DM_host vs log Sigma*	All	51	+0.17 (0.24)	0.64	[-0.13, +0.44]	0.00	+0.11 (0.24)	+0.14 (0.04; 0.01)	0.41	0.84
DM_host vs R_e	All	55	-0.10 (0.45)	0.52	[-0.38, +0.18]	0.00	-0.08 (0.39)	-0.13 (0.03; 0.01)	0.67	0.75
DM_host vs offset / R_e	All	45	+0.17 (0.28)	0.69	[-0.15, +0.44]	0.00	+0.12 (0.21)	+0.12 (0.03; 0.01)	0.45	0.84
DM_host vs A_V (old stars)	All	30	+0.35 (0.055)	0.10	[-0.04, +0.68]	0.29	+0.25 (0.039)	+0.26 (0.15; 0.11)	0.17	0.23
DM_host vs A_V (young stars)	All	31	+0.10 (0.60)	0.53	[-0.30, +0.48]	0.00	+0.07 (0.56)	+0.05 (0.00; 0.00)	0.76	0.75
Host halo										
DM_host vs predicted host-halo DM	All	44	-0.06 (0.72)	0.80	[-0.35, +0.24]	0.00	-0.04 (0.64)	-0.08 (0.01; 0.00)	0.80	0.94
Propagation										
log tau vs DM_host	All	29	+0.45 (0.014)	0.002	[+0.10, +0.71]	0.88	+0.30 (0.016)	+0.38 (0.55; 0.47)	0.12	0.07
log |RM| vs DM_host	All	41	+0.49 (0.001)	0.036	[+0.20, +0.72]	1.00	+0.37 (0.034)	+0.38 (0.77; 0.66)	0.04	0.19
log tau vs log Sigma_SFR	All	23	-0.04 (0.86)	0.097	[-0.48, +0.46]	0.00	+0.03 (0.83)	–	0.92	0.23
log tau vs log Sigma*	All	26	+0.08 (0.68)	0.20	[-0.36, +0.51]	0.00	+0.08 (0.52)	–	0.80	0.37
log tau vs offset / R_e	All	24	-0.08 (0.71)	0.61	[-0.47, +0.34]	0.00	-0.04 (0.82)	–	0.80	0.83
log |RM| vs log Sigma_SFR	All	33	+0.21 (0.24)	0.35	[-0.13, +0.53]	0.00	+0.16 (0.16)	–	0.41	0.59
log |RM| vs offset / R_e	All	33	+0.23 (0.20)	1.00	[-0.16, +0.55]	0.01	+0.19 (0.095)	–	0.38	1.00
linear polarization vs log |RM|	All	23	+0.32 (0.13)	0.13	[-0.09, +0.65]	0.08	+0.23 (0.16)	–	0.27	0.26
Sanity checks										
log SFR vs log M*	All	46	+0.25 (0.097)	0.41	[-0.09, +0.55]	0.12	+0.15 (0.11)	–	–	–
stellar metallicity vs log M*	All	31	+0.27 (0.14)	0.52	[-0.09, +0.55]	0.01	+0.18 (0.12)	–	–	–
R_e vs log M*	All	51	+0.33 (0.018)	0.028	[+0.03, +0.60]	0.90	+0.25 (0.010)	–	–	–
mass-weighted age vs log M*	All	30	+0.38 (0.036)	0.040	[+0.02, +0.66]	0.50	+0.32 (0.007)	–	–	–
A_V (young stars) vs log M*	All	31	-0.10 (0.59)	0.89	[-0.44, +0.24]	0.00	-0.07 (0.51)	–	–	–

BH q: Benjamini–Hochberg over the 34 hypothesis tests (sanity checks excluded), for Spearman and for AD. MC: median r_S over 2,000 draws of DM_{cosmic} and DM_{MW}, and the fractions with Spearman $p < 0.05$ and AD $p < 0.05$.
AD p: two-sample Anderson–Darling test that y (in 'y vs x'; the property, against q) has the same distribution for hosts below and above the median of x (permutation p, about 5,000 shuffles).